

Analysis of Factors Affecting Student Academic Performance

Introduction

In this project, I am exploring how different factors like study hours, attendance, and activities affect student performance.

I am using basic data analysis techniques such as cleaning data, grouping, and simple calculations to understand patterns in the dataset.

The main questions I want to answer are:

- How do study hours affect student performance?
- Do students who study more have fewer absences?
- Which factors seem to have the biggest impact on performance?

The goal is to understand what habits might help students perform better so I will focus on variables such as study hours, absences, extracurricular activities, and part-time jobs to identify patterns and relationships in the data.

Load Data

I am starting with loading and reading data.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset (make sure file is uploaded in Vocareum)
df = pd.read_excel('student-scores.xlsx')

df.head()
```

	id	first_name	last_name	email	gender	pa
0	1	Paul	Casey	paul.casey.1@gslingacademy.com	male	Fa
1	2	Danielle	Sandoval	danielle.sandoval.2@gslingacademy.com	female	Fa
2	3	Tina	Andrews	tina.andrews.3@gslingacademy.com	female	Fa
3	4	Tara	Clark	tara.clark.4@gslingacademy.com	female	Fa
4	5	Anthony	Campos	anthony.campos.5@gslingacademy.com	male	Fa

Data Overview & Cleaning:

I am checking the dataset for missing values and data types. Since there are no missing values, so no major cleaning is needed.

```
# Check structure
```

```
df.info()
```

```
# Check missing values
```

```
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2000 entries, 0 to 1999
```

```
Data columns (total 17 columns):
```

#	Column	Non-Null Count	Dtype
0	id	2000 non-null	int64
1	first_name	2000 non-null	object
2	last_name	2000 non-null	object
3	email	2000 non-null	object
4	gender	2000 non-null	object
5	part_time_job	2000 non-null	bool
6	absence_days	2000 non-null	int64
7	extracurricular_activities	2000 non-null	bool
8	weekly_self_study_hours	2000 non-null	int64
9	career_aspiration	2000 non-null	object
10	math_score	2000 non-null	int64
11	history_score	2000 non-null	int64
12	physics_score	2000 non-null	int64
13	chemistry_score	2000 non-null	int64
14	biology_score	2000 non-null	int64
15	english_score	2000 non-null	int64
16	geography_score	2000 non-null	int64

```
dtypes: bool(2), int64(10), object(5)
```

```
memory usage: 238.4+ KB
```

id	0
first_name	0
last_name	0
email	0
gender	0
part_time_job	0
absence_days	0
extracurricular_activities	0
weekly_self_study_hours	0
career_aspiration	0
math_score	0
history_score	0
physics_score	0
chemistry_score	0
biology_score	0
english_score	0
geography_score	0

```
dtype: int64
```

I am removing columns like names and email because they are not useful for analysis.

```
# Drop unnecessary columns
df = df.drop(columns=['id', 'first_name', 'last_name', 'email'])

df.head()
```

	gender	part_time_job	absence_days	extracurricular_activities	weekly_self
0	male	False	3	False	27
1	female	False	2	False	47
2	female	False	9	True	13
3	female	False	5	False	3
4	male	False	5	False	10

Compute a Derived Column:

I am creating an average score to represent overall academic performance across all subjects.

```
# Create overall performance metric
df['average_score'] = df[
    ['math_score', 'history_score', 'physics_score',
     'chemistry_score', 'biology_score', 'english_score', 'geography_score']
].mean(axis=1)

df.head()
```

	gender	part_time_job	absence_days	extracurricular_activities	weekly_self
0	male	False	3	False	27
1	female	False	2	False	47
2	female	False	9	True	13
3	female	False	5	False	3
4	male	False	5	False	10

Exploratory Analysis

First, I want to answer some basic questions and perform some basic analysis to explore and understand the data.

Which subjects do students perform best and worst in?

Interpretation: From the results, it looks like students score slightly higher in math and geography compared to some other subjects. The differences are not very large, but we can still see some variation across subjects.

```
df[['math_score', 'history_score', 'physics_score', 'chemistry_score', 'biology_score', 'english_score', 'geography_score']].mean()
```

```
math_score      83.4520
history_score    80.3320
physics_score    81.3365
chemistry_score  79.9950
biology_score    79.5815
english_score    81.2775
geography_score  80.8880
dtype: float64
```

Do students perform consistently across subjects?

Interpretation: The correlations between subjects are positive but weak, so it seems that performance in one subject does not strongly predict performance in others.

```
df[['math_score', 'physics_score', 'chemistry_score']].corr()
```

	math_score	physics_score	chemistry_score
math_score	1.000000	0.115719	0.127131
physics_score	0.115719	1.000000	0.126163
chemistry_score	0.127131	0.126163	1.000000

Do students with part-time jobs study less?

Interpretation: Students with part-time jobs seem to study a little less on average. This might be because they have less time available.

```
df.groupby('part_time_job')['weekly_self_study_hours'].mean()
```

```
part_time_job
False      18.879454
True       11.765823
Name: weekly_self_study_hours, dtype: float64
```

Do extracurricular activities reduce study time?

Interpretation: For extracurricular activities, the difference in study time is not very large, so the impact is not very clear.

```
df.groupby('extracurricular_activities')['weekly_self_study_hours'].mean()
```

```
extracurricular_activities
False    18.077889
True     16.497549
Name: weekly_self_study_hours, dtype: float64
```

Do male and female students perform differently?

Interpretation: The results show that male students have a slightly higher average score compared to female students. However, the difference is small, so overall performance between genders is fairly similar.

```
df.groupby('gender')['average_score'].mean()
```

```
gender
female    80.894069
male      81.066991
Name: average_score, dtype: float64
```

Do gender differences vary by subject?

Interpretation: There are small differences between male and female students across subjects. For example, one group seems to perform slightly better in math while the other does slightly better in English. However, these differences are not very large, so performance is fairly balanced across genders.

```
df.groupby('gender')[['math_score', 'english_score']].mean()
```

	math_score	english_score
gender		
female	82.792415	81.740519
male	84.114228	80.812625

Additional Exploratory Analysis

Is there a “threshold” where more study hours stop helping?

Interpretation: As the study hours increase, average scores also increase, but the improvement is not consistent at higher levels. After a certain point, the gains become smaller, suggesting that studying more does not always lead to significantly better results.

```
df.groupby('weekly_self_study_hours')['average_score'].mean()
```

```
weekly_self_study_hours
```

```
0    76.357143
1    76.991974
2    77.134831
3    78.070028
4    76.752101
5    77.022796
6    80.322981
7    80.298701
8    79.086735
9    79.489011
10   80.742857
11   81.011583
12   80.635338
13   81.285714
14   81.418182
15   81.168203
16   80.633053
17   82.277551
18   79.562500
19   80.533835
20   81.750000
21   80.467532
22   83.206349
23   81.556777
24   81.623377
25   81.130699
26   81.482436
27   82.414286
28   82.259740
29   81.167070
30   83.180758
31   82.491837
32   81.986813
33   83.417582
34   82.944700
35   92.303297
36   93.142857
37   92.785714
38   92.047619
39   91.428571
40   91.095238
41   91.892857
42   92.591837
43   92.333333
44   93.904762
46   92.523810
47   92.160714
48   93.400000
49   92.821429
50   90.928571
```

```
Name: average_score, dtype: float64
```

Do students with very high absences perform significantly worse?

Interpretation: Initially, I tried to analyze students with more than 10 absence days, but the dataset did not contain any such cases. To resolve this, I adjusted the threshold using the 75th percentile of absence days, which allowed for a more meaningful comparison between high and low absence groups. This adjustment ensures that both groups contain enough data points for a meaningful comparison.

```
threshold = df['absence_days'].quantile(0.75)

high_absence = df[df['absence_days'] > threshold]['average_score'].mean()
low_absence = df[df['absence_days'] <= threshold]['average_score'].mean()

high_absence, low_absence

(78.27703749612644, 81.79012345679013)
```

Who are the top-performing students?

Interpretation: The top-performing students are those with the highest average scores in the dataset. These students generally appear to have higher study hours and lower absence days compared to others.

```
df.sort_values(by='average_score', ascending=False).head(10)
```

	gender	part_time_job	absence_days	extracurricular_activities	weekly_
796	female	False	3	True	35
779	male	False	6	False	35
1573	female	True	2	False	48
551	male	False	10	False	35
1419	female	False	1	False	35
1122	male	False	2	False	35
163	male	False	1	True	44
985	male	False	2	True	35
718	female	False	3	False	35
541	male	False	2	False	35

What do top students have in common?

Interpretation: Top students tend to study more and miss fewer classes, which suggests that both effort and consistency play an important role in achieving higher scores.

```
top_students = df[df['average_score'] > df['average_score'].quantile(0.75)]  
  
top_students[['weekly_self_study_hours', 'absence_days']].mean()
```

```
weekly_self_study_hours    25.470588  
absence_days               3.221095  
dtype: float64
```

Core Analysis

In this section, I am focusing on answering the key questions defined earlier using structured analysis and visualizations.

Question 1:

How do study hours affect student performance?

Interpretation: From the results, average scores generally increase as weekly study hours increase. However, the increase is not perfectly consistent across all study hours. Some values go slightly up or down, which means that study time is important, but it is not the only factor affecting performance.

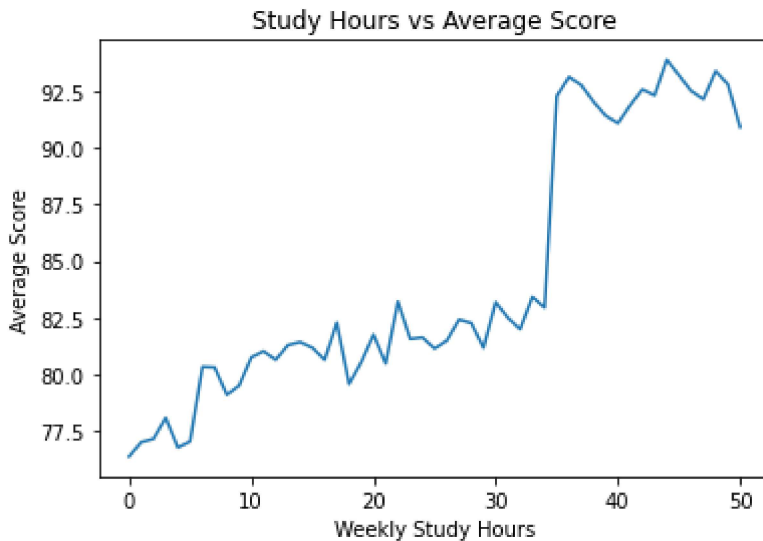
```
study_performance = df.groupby('weekly_self_study_hours')['average_score'].mean()  
study_performance.head()
```

```
weekly_self_study_hours  
0    76.357143  
1    76.991974  
2    77.134831  
3    78.070028  
4    76.752101  
Name: average_score, dtype: float64
```

Study Performance Visualization

Interpretation: Students who spend more time studying generally achieve higher scores, which is expected. That said, the improvement isn't perfectly steady across all study hours. After a certain point, the gains appear smaller, suggesting that simply increasing study time may not always lead to proportionally better results.

```
study_performance.plot(kind='line')
plt.title("Study Hours vs Average Score")
plt.xlabel("Weekly Study Hours")
plt.ylabel("Average Score")
plt.show()
```



Question 2:

Do students who study more have fewer absences?

Interpretation: Students who study more hours seem to have fewer absences. The number of absence days goes down slightly as study hours increase, but the change is small. This means there may be a relationship, but it is not very strong.

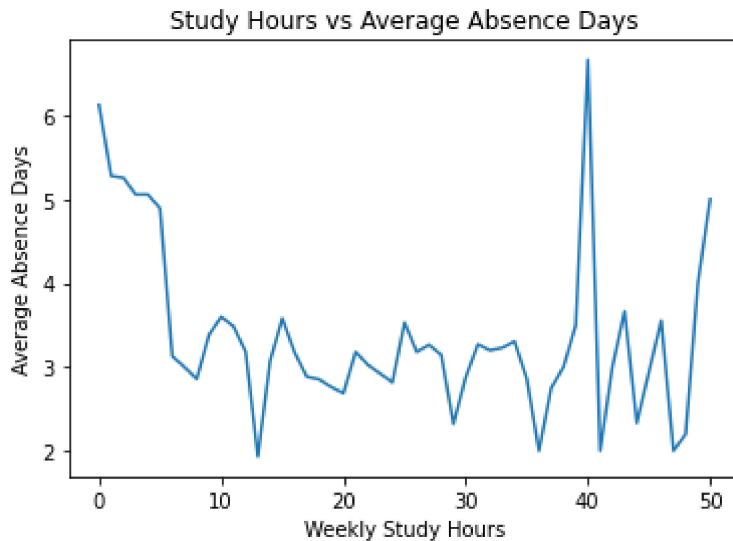
```
study_absence = df.groupby('weekly_self_study_hours')['absence_days'].mean()
study_absence.head()
```

```
weekly_self_study_hours
0    6.128571
1    5.280899
2    5.258427
3    5.058824
4    5.058824
Name: absence_days, dtype: float64
```

Study Hours vs Absence Days Visualization

Interpretation: Students who study more hours seem to have fewer absences. The relationship is not perfectly clear, but it shows that students who study more may also attend classes more regularly. This suggests that study habits and attendance may be related.

```
study_absence.plot(kind='line')
plt.title("Study Hours vs Average Absence Days")
plt.xlabel("Weekly Study Hours")
plt.ylabel("Average Absence Days")
plt.show()
```



Question 3:

Which factor has the greatest impact on performance?

- To better understand which factors matter most, I compare their overall impact on performance using the variation in average scores.

Interpretation: Study hours have the biggest impact on performance, since they show the largest difference in average scores. Absence days also affect performance, but not as much. Part-time jobs have a smaller effect, and extracurricular activities have very little impact. This means study time is the most important factor among the ones analyzed.

```
# Calculate impact as range of average scores across each factor
```

```
study_impact = df.groupby('weekly_self_study_hours')['average_score'].mean()  
absence_impact = df.groupby('absence_days')['average_score'].mean()  
job_impact = df.groupby('part_time_job')['average_score'].mean()  
activity_impact = df.groupby('extracurricular_activities')['average_score'].mean()
```

```
factors = {  
    'Study Hours': study_impact.max() - study_impact.min(),  
    'Absence Days': absence_impact.max() - absence_impact.min(),  
    'Part-Time Job': job_impact.max() - job_impact.min(),  
    'Extracurricular Activities': activity_impact.max() - activity_impact.min()  
}
```

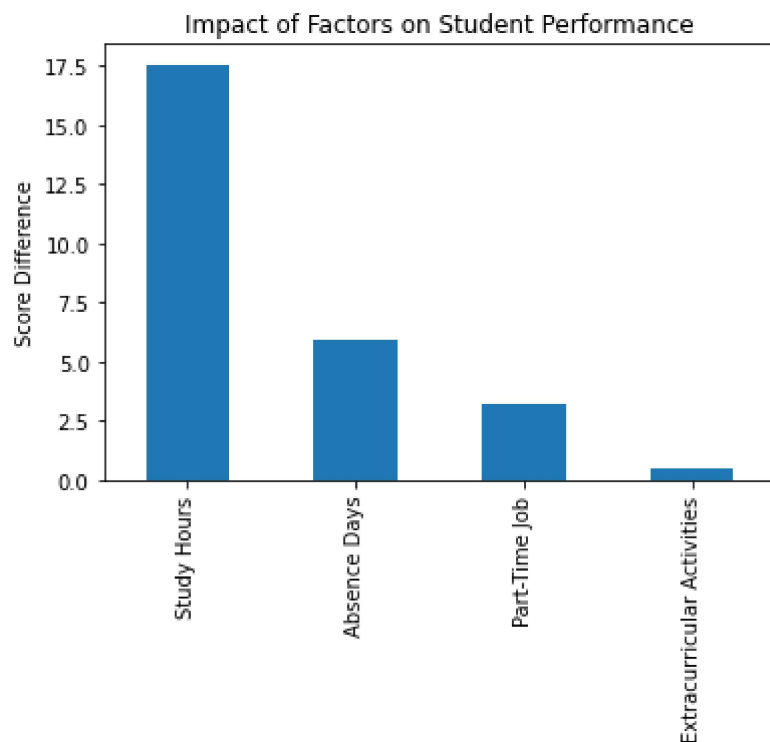
```
factor_impact = pd.Series(factors)  
factor_impact
```

```
Study Hours          17.547619  
Absence Days         5.863520  
Part-Time Job        3.193845  
Extracurricular Activities  0.478979  
dtype: float64
```

Factors Impact Visualization

Interpretation: Study hours have the biggest impact on performance. Absence days also have an effect, but not as much. Part-time jobs and extracurricular activities do not seem to make a big difference.

```
factor_impact.plot(kind='bar')  
plt.title("Impact of Factors on Student Performance")  
plt.ylabel("Score Difference")  
plt.show()
```



Key Insights & Summary

- Students who study more hours per week tend to achieve higher scores.
- Increased absence days are associated with lower academic performance.
- Having a part-time job may slightly impact academic outcomes, possibly due to time constraints.
- Extracurricular activities show varied effects, indicating that balance is important.

Advanced Analysis: Predicting Student Performance

In addition to exploratory analysis, I am applying a simple predictive model to better understand how study habits influence academic performance. Specifically, I am using linear regression to examine whether weekly study hours can be used to predict a student's average score.

Linear Regression Model

Linear Regression Model

In this step, I use linear regression to understand how study hours affect average score. The goal is to see if study time can help predict student performance and to estimate how much scores change as study hours increase.

Interpretation: The results show a positive relationship between study hours and average score. This means that students who study more generally tend to score slightly higher.

The slope value (about 0.25) suggests that each additional hour of study is linked to a small increase in score. However, the effect is not very large, which means other factors also influence performance. The intercept (around 76.5) represents the expected score when study hours are zero, but this may not be very meaningful in real situations. Overall, this supports the earlier analysis that study time has an impact on performance, but it is not the only factor.

```
import numpy as np

# Extracting the independent variable (X): weekly study hours
# .values converts the column into a NumPy array for numerical computation
X = df['weekly_self_study_hours'].values

# Extracting the dependent variable (y): average score
# This is what we are trying to predict using study hours
y = df['average_score'].values

# Using NumPy's polyfit to compute the best-fit line (linear regression)
# Degree = 1 means a straight line → returns slope (m) and intercept (b)
m, b = np.polyfit(X, y, 1)

# Generating predicted values using the regression equation:  $y = mX + b$ 
# This represents the estimated scores based on study hours
y_pred = m * X + b

# Displaying the model parameters
# Slope (m): change in score per additional study hour
# Intercept (b): estimated score when study hours = 0
print("Slope:", m)
print("Intercept:", b)
```

Slope: 0.24982646933106756

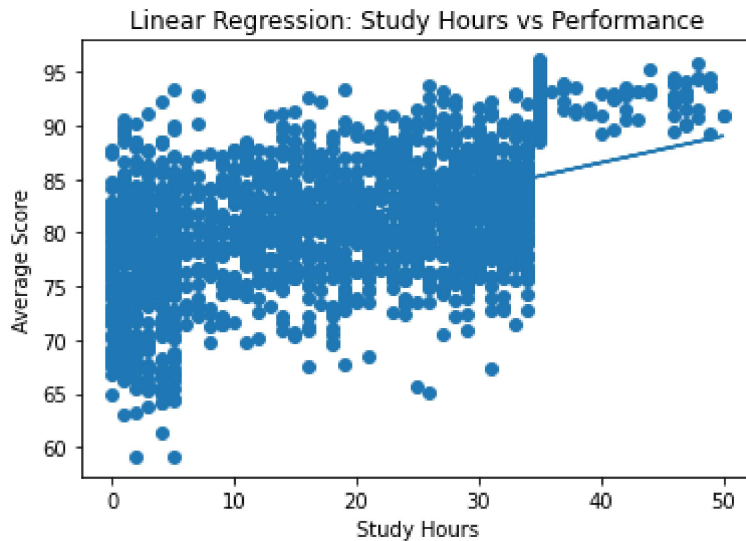
Intercept: 76.5445632666494

Linear Regression Visualization

Interpretation: The graph shows that as study hours increase, average scores also increase. The relationship is not perfect, so other factors also affect performance. But overall, studying more seems to help.

```
import matplotlib.pyplot as plt

plt.scatter(X, y)
plt.plot(X, y_pred)
plt.title("Linear Regression: Study Hours vs Performance")
plt.xlabel("Study Hours")
plt.ylabel("Average Score")
plt.show()
```



Correlation Check:

I am using a correlation matrix next, to examine relationships between numerical variables. This will help in identifying which factors are most strongly associated with student performance.

Interpretation: Study hours are positively related to average score and this means that study time and attendance are important factors for performance. Absence days are negatively related to performance, so students who miss more classes tend to score lower. Most subjects are also related to each other, meaning students who do well in one subject often do well in others. This shows that study time and attendance are important factors for performance.

```
numeric_df = df.select_dtypes(include=['number'])
numeric_df.corr()
```

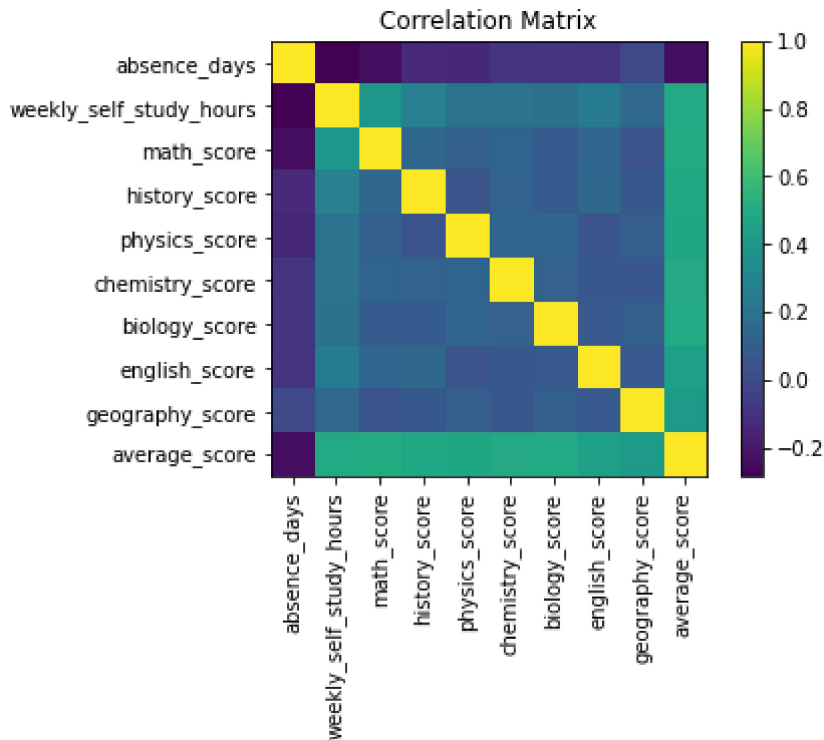
	absence_days	weekly_self_study_hours	math_sco
absence_days	1.000000	-0.286086	-0.237072
weekly_self_study_hours	-0.286086	1.000000	0.393569
math_score	-0.237072	0.393569	1.000000
history_score	-0.128159	0.276231	0.147247
physics_score	-0.136419	0.202120	0.115719
chemistry_score	-0.084029	0.201340	0.127131
biology_score	-0.090554	0.190481	0.081298
english_score	-0.084861	0.247796	0.134831
geography_score	-0.002942	0.153622	0.049672
average_score	-0.232868	0.501520	0.508116

Correlation Visualization

Interpretation: The heatmap shows how different variables are related. Lighter colors mean a stronger relationship, and darker colors mean a weaker relationship. Study hours are positively related to performance, while absence days are negatively related. This supports the earlier results that studying more and attending classes regularly help improve performance.


```
corr_matrix = numeric_df.corr()

plt.imshow(corr_matrix)
plt.colorbar()
plt.title("Correlation Matrix")
plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns, rotation=90)
plt.yticks(range(len(corr_matrix.columns)), corr_matrix.columns)
plt.show()
```



Conclusion

Overall, this analysis shows that study habits and attendance are the strongest factors influencing student performance. Students who consistently study and attend classes tend to achieve better results. While part-time jobs and extracurricular activities also play a role, their impact appears to depend more on how well students manage their time.

- Students who study more hours per week generally achieve higher scores.
- Higher absence days are associated with lower academic performance.
- Part-time jobs may have a small impact, likely due to time constraints.
- Extracurricular activities show mixed effects, suggesting that balance is important.

Limitations

This analysis is based on observable data, so it does not establish causation. For example, while study hours are associated with higher performance, other factors such as student motivation or prior knowledge may also influence outcomes. Additionally, the dataset may not capture all variables that affect academic performance.